



Research paper

Pyrogenicity of phage active pharmaceutical ingredients used for personalized therapy in Belgium

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ABSTRACT

The assessment of pyrogenicity is critical for phage-based therapeutic products, especially when administered intravenously. However, current standard endotoxin tests, such as the *Limulus* Amoebocyte Lysate (LAL) and the recombinant Factor C (rFC) assay, capture only part of their immune-activating potential. Here, we compared the LAL test, the rFC assay, and the human cell-based Monocyte Activation Test (MAT) to evaluate the pyrogenic profile of phage active pharmaceutical ingredients (pAPIs). While LAL and rFC are rapid and cost-effective, MAT provides unique physiological relevance by capturing immune responses to both endotoxins and non-endotoxin pyrogens (NEPs), as well as potential interactions between phages and the human immune system. This study provides a first insight into the pyrogenic activity of pAPIs as perceived by human immune cells and highlights the complementary roles of endotoxin tests and MAT in ensuring product safety. The tested pAPIs generally exhibited low pyrogenicity, with most batches below 1000 EU/mL. Based on our experience, we propose a practical two-tiered quality control strategy using rFC as an initial screening tool for rapid and cost-effective endotoxin quantification, and MAT as a complement to assess pyrogenic potential as perceived by the human immune system. Importantly, MAT is not intended to overrule endotoxin detection by rFC, but rather to support the interpretation of complex pyrogenic signals. This approach provides a more physiologically relevant and comprehensive view of pyrogenicity, while supporting ethical testing practices aligned with the 3Rs principles (Replacement, Reduction, and Refinement of animal use).

1. Introduction

Antimicrobial resistance has become a major public health concern, with an global annual death toll of more than 1 million. By the year 2050, this grim count is expected to increase to 10 million if no measures are taken [1]. As existing antimicrobials are starting to fail, and few new antimicrobials are being approved for clinical use [2], the pharmaceutical landscape is in dire need of alternative treatments. In this context, the use of bacteriophages (phages) to target pathogenic bacteria presents a promising adjunct approach to address difficult-to-treat

infections. Phages are viruses that specifically recognize and kill their bacterial host, making them a powerful ally against bacterial infection [3]. Over the past decades, considerable effort has been invested in the development of industrial phage therapy medicinal products under good manufacturing practices [4]. These products are typically designed as off-the-shelf medicines, often formulated as phage cocktails to target a broad range of bacterial strains [5,6]. While fixed phage cocktails can provide ready-to-use treatments, their use has sometimes been associated with challenges in demonstrating efficacy in randomized controlled trials [7,8]. In contrast, personalized, tailor-made magistral phage

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preparations make it possible to select phages that infect a patient's specific bacterial isolate, and even pre-adapt them – using directed evolution – to increase their activity and lower bacterial resistance progression [9]. These personalized products are typically prepared by hospitals or compounding pharmacies using phage active pharmaceutical ingredients (pAPIs), which serve as the basis for individualized formulations. In Belgium, personalized phage preparations have already been used in over 100 consecutive cases, with clinical improvement and eradication reported in 77.2% and 61.0% of cases, respectively [10]. The regulatory challenges associated with personalized phage medicines are considerable, and often conflict with the time constraints linked to the urgency of therapy for critically ill patients. Since Belgium has implemented personalized phage therapy under a magistral framework since 2018, with QC performed by our Official Medicines Control Laboratory (OMCL), substantial experience has been gained. The current QC procedures in place at the Belgian OMCL for the control of pAPIs are illustrated in Fig. 1.

In January 2025, a novel general chapter on phage therapeutics was published in the European Pharmacopoeia (Ph. Eur.), providing a harmonised European framework of requirements for the production and control of phage therapy active substances and medicinal products (see Supplementary Fig. S1). While the general chapter is published for information, it suggests some key quality control (QC) criteria that should be implemented to ensure patient safety.

A critical concern for the safety of pAPIs, especially when produced or pre-adapted on clinical isolates, is their pyrogenicity, the potential of a substance to induce an inflammatory response upon introduction into the body. Phage products have a wide range of potential therapeutic applications and can be administered through multiple routes depending on the infection type, location, and therapeutic objectives [11]. Especially intravenous administration poses a heightened risk, as it directly introduces phages (and any potential pyrogenic contaminants) into the bloodstream, where they can provoke systemic inflammation or adverse immune reactions [12].

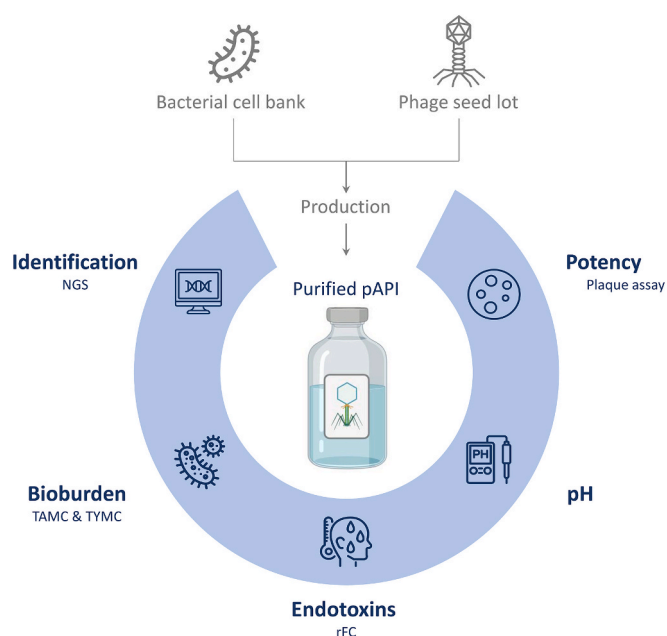


Fig. 1. Quality control testing of phage active pharmaceutical ingredients (pAPIs) at the Official Medicines Control Laboratory (OMCL) in Belgium. Current assays include sequencing for phage identification, plaque assay for potency, recombinant Factor C (rFC) assay for endotoxin levels, microbial contamination checks (total aerobic microbial count [TAMC] and total yeast and mold count [TYMC]), and pH measurement. The human cell-based Monocyte Activation Test (MAT) for pyrogenicity testing is currently being implemented.

The most prevalent and potent type of pyrogens are bacterial endotoxins, lipopolysaccharides (LPS) of the outer membrane of Gram-negative bacteria [13]. Other pyrogens, grouped under the term “non-endotoxin pyrogens” (NEPs), encompass a wide variety of components of bacterial, fungal, or viral origin, as well as environmental contaminants that can also trigger fever. Because phage pre-adaptation and production rely on propagation in bacterial hosts, which can be either Gram-negative or Gram-positive and are highly diverse due to the specific nature of phages, the risk of contamination of pAPIs with both endotoxins and NEPs is substantial [14].

The Rabbit Pyrogen Test (RPT), which measures rabbit body temperature after intravenous injection of a test sample, has long been the gold standard for detecting pyrogenicity in parenteral products [15]. Due to ethical concerns, limited sensitivity, poor human relevance, and in line with the 3Rs principles (Replacement, Reduction, Refinement), all references to the RPT were removed from the Ph. Eur. on the 1st of July 2025 [16]. Simultaneously, the new general chapter 5.1.13 “Pyrogenicity” was published, providing guidance for selecting and implementing suitable tests based on thorough risk assessment [17,18]. Table 1 provides a comparison of the different methods.

The *Limulus* Amoebocyte Lysate (LAL) test, also known as the Bacterial Endotoxin Test (BET), was developed in the 1970 s as an *in vitro* method for products where endotoxins are expected to be the only source of pyrogenicity [19]. Although the LAL test avoids mammalian testing, it remains animal-dependent, relying on the clotting cascade of horseshoe crab (*Limulus polyphemus*) blood [20]. Harvesting this blood through capture, bleeding, and release raises ethical and ecological concerns, given the species' vulnerability and vital role in marine ecosystems [21].

Recent advances have led to the development of the recombinant Factor C (rFC) assay, a synthetic, animal-free method comparable to the LAL test and now included in compendial guidelines [22]. The rFC assay employs a recombinant form of the horseshoe crab's Factor C enzyme, the first and most critical component of the clotting cascade in the LAL test, produced through genetic engineering. By reducing the system to a single reaction, the rFC assay avoids the complexity of the native LAL cascade and its susceptibility to interference. In particular, it is unaffected by β -glucans and other factors that can trigger false positives or inhibit downstream proteases in the LAL system, thus providing a more specific and reliable method for bacterial endotoxin detection [19].

Nevertheless, because the rFC assay relies on the same primary recognition mechanism as LAL (activation of Factor C), both methods share the inherent limitation of being vulnerable to endotoxin masking, also referred to as low endotoxin recovery (LER). In this phenomenon, certain drug formulations, particularly those containing chelating agents and non-ionic surfactants, disrupt the supramolecular aggregates of endotoxin, rendering it undetectable by these assays while it remains biologically active [23].

The most recent evolution in compendial pyrogenicity testing is the Monocyte Activation Test (MAT), which measures the release of pro-inflammatory cytokines by human monocytes upon pyrogen exposures [24]. Unlike LAL and rFC, which only determine endotoxin content, MAT provides a comprehensive, human-relevant assessment of pyrogen contamination, including synergistic pyrogenic effects [25,26].

Several MAT approaches exist, differing primarily in the source of monocytes. The whole blood assay utilizes freshly drawn human blood, which may be cryopreserved [27], preserving the natural plasma environment for physiologically relevant responses [28], but it faces limitations due to donor availability and variability, as well as complex regulatory and shipment conditions. The cell line-based assay, typically using MonoMac-6 cells, offers standardized testing but may not fully replicate the broad responsiveness of primary human monocytes, particularly to certain NEPs [29]. The PBMC-based MAT uses peripheral blood mononuclear cells (PBMCs), offering a balance between physiological relevance and standardization, making it suitable for routine testing [28,30].

In this study, we compared the outcomes of the LAL, rFC, and MAT assays applied to PAPs, intended for the magistral preparation of personalized phage therapeutics, produced in both Gram-positive and Gram-negative bacteria.

2. Materials & methods

The LAL and rFC kits were selected based on their availability and their established use for routine vaccine testing within our laboratory, ensuring methodological consistency with ongoing quality control procedures and facilitating comparability with previously generated data.

The MAT approach was selected mainly based on practical considerations. Obtaining sufficient quantities of whole blood from healthy volunteers for routine MAT testing poses significant logistical challenges. Furthermore, previous experience with cryopreserved whole blood has revealed issues related to its biological activity, which is highly sensitive to transport conditions [31]. Cell lines such as MonoMac-6 require continuous culturing, which introduces risks of genetic drift and contamination, and they fail to fully replicate the broad immunological responsiveness of primary monocytes. Therefore, we opted to use commercially available, cryopreserved PBMCs derived from a pooled sample of four donors, supplied with a certificate of analysis from the manufacturer. This approach ensures reproducibility, standardization, and compliance with quality requirements while allowing us to implement the MAT reliably in our setting. Two commercial kits were evaluated during assay qualification, and the kit described below was selected based on its superior sensitivity and reproducibility. IL-6 ELISA was chosen as the read-out method because it exhibits higher sensitivity compared to interleukin-1 beta (IL-1 β) and tumor necrosis factor α (TNF- α) [30], is secreted in larger quantities into the cell-conditioned medium [32], and because this was the assay recommended by the supplier of the PBMCs.

2.1. *Limulus* Amoebocyte Lysate (LAL) test

The LAL test was performed using a chromogenic kinetic LAL assay from Lonza (Kinetic-QCL®) (Basel, Switzerland). This kit is fully validated following the Ph. Eur. guidelines [33,34] and has been used in our official medicines control laboratory (OMCL) to test vaccines for several years. Briefly, reference standard endotoxin from the kit was reconstituted with pyrogen-free water to obtain a concentration of 50 endotoxin units per milliliter (EU/mL), then it was vortexed for 30 min to ensure sufficient homogenization. Ten-fold dilutions were subsequently prepared with pyrogen-free water in pyrogen-free glass tubes, vortexing each one for at least one minute, to obtain the five points for the calibration curve (ranging from 0.005 EU/mL to 50 EU/mL). Samples were diluted 1:1000 in pyrogen-free water to mitigate potential matrix effects that could compromise assay validity, while preserving a detection sensitivity of 5 EU/mL, as defined by the lowest standard concentration. When this dilution did not appear to be sufficient (measurement above the highest point of the standard curve or invalid spike recovery), the sample was further diluted and tested again. For all standards and samples, technical duplicates were tested, and their mean was used for calculations. To exclude the presence of an interfering effect of the sample matrix, per sample, two additional replicates were spiked to a final concentration of 0.5 EU/mL. A recovery within the range of 50% to 200% was required for the result to be valid, as per Ph. Eur. guidelines [33]. Finally, per test, two positive controls containing a known endotoxin concentration were included as samples. Following the manufacturer's protocol, each plate was pre-incubated for 15 min at 37°C, before adding the kinetic reagent (containing the lysate and the chromogenic substrate) previously reconstituted with pyrogen-free water. Thereafter, the wells were monitored over time for the appearance of a yellow color by a spectrophotometer (Agilent BioTek Epoch 2) set at 405 nm and steered by the BioTek Gen5

Table 1

Comparative overview of compendial methods for the detection of pyrogens and/or endotoxins, including the Rabbit Pyrogen Test (RPT), the *Limulus* Amoebocyte Lysate (LAL) assay, the recombinant Factor C (rFC) assay, and the Monocyte Activation Test (MAT). LAL, rFC, and MAT were applied in this study. Ph. Eur. = European Pharmacopoeia; USP = United States Pharmacopeia; JP = Japanese Pharmacopoeia.

	RPT	LAL	rFC	MAT
Principle	Measures rise in body temperature in rabbits after injection of test sample	Detects endotoxins via coagulation cascade from horseshoe crab blood	Recombinant Factor C (initial step of LAL cascade) activation upon endotoxin binding	Detects cytokine release (e.g. IL-6) from human monocytes upon pyrogen exposure
Detection scope	Broad: endotoxins + non-endotoxin pyrogens (all fever-inducing agents)	Gram-negative bacterial endotoxins (LPS) only	Gram-negative bacterial endotoxins (LPS) only	Broad pyrogen detection (endotoxins + non-endotoxin pyrogens such as lipoteichoic acid, peptidoglycan, viral components...)
Biological source	Live rabbits (in vivo animal test)	Horseshoe crabs (animal-derived, ecological concerns)	Recombinant protein (animal-free, sustainable)	Human whole blood or PBMCs (ex vivo, donor-dependent)
Ethical concerns	Very high (animal suffering, invasive test)	Significant (harvesting harms crab populations)	Low (no animals used)	Moderate (requires human blood, donor variation, ethical approvals)
Specificity	Physiologically broad but non-specific (any pyrogen causing fever)	High for endotoxin, but no detection of non-endotoxin pyrogens. Subject to interference by β -glucans.	High for endotoxin, but no detection of non-endotoxin pyrogens	Physiologically relevant—detects both endotoxin and non-endotoxin pyrogens
Sensitivity	Relatively low (μ g/mL range; less sensitive than <i>in vitro</i> methods)	Very high (down to ~ 0.005 EU/mL)	Comparable to LAL (very high)	Lower than LAL/rFC for endotoxin (down to ~ 0.05 EE/mL); varies with donor
Reproducibility	Variable (animal-to-animal variability, ethical restrictions limit use)	High, standardized	High, standardized	Moderate—affected by donor-to-donor variability, higher with pooled blood
Relevance to humans	Indirect (animal physiology, not fully representative)	Indirect (horseshoe crab biology, not human immune system)	Indirect (engineered protein, not human biology)	Direct (human monocytes reflect actual immune response)
Applications	Historical compendial test for pyrogens; largely obsolete	Widely used regulatory standard for endotoxin testing in pharmaceuticals	Increasingly accepted alternative to LAL; regulatory support growing	Human cell-based pyrogen test (especially for vaccines, biologics, cell/gene therapies)
Regulatory status	Still in some pharmacopoeias (e.g., USP) but largely replaced; last-resort method	Compendial standard (USP, Ph. Eur., JP); still dominant globally	Compendial method in Ph. Eur. and USP; regulatory recognition growing	Compendial method in Ph. Eur. (replacing RPT); accepted alternative to RPT in USP; less widespread outside EU
Throughput	Very low throughput, time- and labor-intensive	High throughput, easy automation	High throughput, easy automation	Lower throughput; requires fresh/donor blood or cryopreserved cells
Cost	High (animal housing, care, ethical overhead)	Cost-effective	Comparable to LAL	Higher cost than LAL and rFC, more economical than RPT

software. A linear standard curve was generated using the $\log_{10}$ reaction time (time to reach a 0.2 unit increase in absorbance) and the $\log_{10}$ endotoxin concentration of the five standards, and subsequently used to infer the samples' endotoxin concentration.

2.2. Recombinant factor C (rFC) assay

For the recombinant factor C (rFC) test, bioMérieux's ENDOZYME® II GO kit (Marcy-l'Étoile, France) was used, which is also fully validated according to Ph. Eur. guidelines [35,36]. In this kit, each 96-well plate is pre-coated with a standard curve (0.005 EU/mL to 50 EU/mL) and spikes (0.5 EU/mL final concentration). It is a fluorescence end-point assay involving a recombinant version of the Factor C enzyme (found in LAL) and a synthetic fluorogenic substrate. Similar to the LAL test, samples were diluted 1000-fold in pyrogen-free water, and further diluted and retested if they were above the quantification range or if spike recovery was invalid. Again, all samples and standards were run in duplicate and the mean was used in calculations. As per the manufacturer's protocol, the coated standard curve was resolubilized by adding 100 μ l pyrogen-free water to the wells, whereas 100 μ l of diluted sample was added to empty and spiked wells. The plate was pre-incubated for 5 min at 37°C, and then 100 μ l of reagent (containing the rFC enzyme and the substrate) was added. The fluorescence was measured at time point zero after a short 15-second shake at an excitation wavelength of 380 nm and an emission wavelength of 440 nm, and again after 60 min of incubation at 37°C in the BioTek Synergy HTX multimode microplate reader (Agilent, Santa Clara, CA, USA) coupled to the BioTek Gen5 software. The difference between the readings was calculated and corrected with the fluorescence obtained for the blank. The $\log_{10}$ of this value and the $\log_{10}$ endotoxin concentration of the standards were used to generate a linear standard curve between 0.005 and 5 EU/mL, from which sample concentrations were inferred. As with the LAL assay, spiked sample recovery was required to fall within the range of 50% to 200%.

2.3. Monocyte activation test (MAT)

The Monocyte Activation Test (MAT) was performed with CTL-MAT (Cleveland, Ohio) kits bought from MAT Biotech (Abcoude, The Netherlands) using cryopreserved human pooled PBMCs from four donors, as a source of monocytes. Following the manufacturer's protocol, the cells were co-cultured with the samples in a humidified incubator at 37°C in the presence of 5% CO₂ for 19 h, prior to measuring cytokine production in the supernatant with an ELISA targeting human interleukin-6 (IL-6). The *Escherichia coli* LPS reference standard obtained with the kit was used at concentrations ranging from 0.016 EU/mL to 1 EU/mL to construct a four-parameter logistic calibration curve, allowing the interpolation of sample pyrogenicity in endotoxin-equivalent units per mL (EE/mL). Samples were diluted in culture medium to three different concentrations based on the results obtained with LAL and rFC to ensure at least one response would fall within the narrow quantification range of the assay. To confirm the test detected pyrogenicity originating from both endotoxins (LPS) and non-endotoxin pyrogens (NEPs), and to check for interference in the test, samples were spiked separately with a known concentration of LPS (0.1 EU/mL) and a known concentration of heat-killed *Staphylococcus aureus* (HKSA, CTL-MAT; 2×10^6 cells/mL). A recovery between 50% and 200% was required for the LPS spike, while for NEPs, a minimum recovery of 50% was accepted, reflecting the inherently variable and sometimes synergistic response to these substances [26]. Higher recoveries, which are less predictable than those observed with LPS, are not penalized. All samples were tested in quadruplicate, following the manufacturer's instructions and in accordance with the requirements of the Ph. Eur. for the MAT method [37].

2.4. Cell viability assay (MTS)

Cell viability was assessed using the CellTiter 96® AQueous One Solution Cell Proliferation Assay (MTS; Promega) according to the manufacturer's instructions, after exposure of the PBMCs to the pAPI samples under the same conditions as used for the MAT. Absorbance was measured at 490 nm with a reference wavelength of 630 nm using a microplate reader (SPECTROstar Nano, BMG Labtech).

Cell viability was expressed as a percentage relative to the negative control (cells cultured in medium only), which was set to 100%. Each condition was tested in nine technical replicates. Statistical analysis was performed using one-way ANOVA followed by Dunnett's multiple comparisons test versus the negative control, with $p < 0.05$ considered significant.

2.5. Phage samples

Phage production samples, i.e., phage active pharmaceutical ingredients (pAPIs), were obtained from the Queen Astrid Military Hospital (QAMH) (Neder-over-Heembeek, Belgium), currently the only producer of pAPIs in Belgium, and selected based on the availability of sufficient volume. Table 2 shows an overview of the phages involved in this study, together with their bacterial hosts, and their titer in plaque-forming units per milliliter (PFU/mL). Samples were produced as described by Pirnay et al. (2024) and stored in airtight tubes at 4°C until use. Aliquots from a single original tube were tested in LAL, rFC, and MAT assays, all conducted within the same month to minimize variability.

3. Results

3.1. Linear range

The LAL and rFC assays, both designed for endotoxin detection, provide results expressed as endotoxin units per milliliter (EU/mL), based on standard curves generated using *E. coli* LPS. In contrast, the MAT assesses overall pyrogenic activity and reports results in endotoxin-equivalent units per milliliter (EE/mL). This is achieved by comparing the cytokine response of human monocytes to the test samples against the response to a standard curve constructed from serial dilutions of *E. coli* LPS. Importantly, the three assays generate standard curves of different shapes and ranges. The LAL test provided the broadest quantification range, from 0.005 to 50 EU/mL, using a linear regression of log-transformed responses (kinetic onset time of color development) against log-transformed endotoxin concentrations. The rFC assay yielded a linear standard curve from 0.005 to 5 EU/mL, based on a regression of the response in relative fluorescence units (net dRFU) against endotoxin concentration. At the highest concentration (50 EU/mL), signal saturation occurred, and we chose not to adjust the gain to avoid compromising sensitivity at the lower end of the curve, as the existing range was sufficient. For the MAT, a seven-point standard curve was modeled with a four-parameter logistic regression, and quantification was performed in the linear portion of the curve, typically between 0.03125 and 0.25 EE/mL, with slight assay-to-assay variation. For this study, samples with results below the limit of quantification were assigned the value of the limit of quantification for analysis.

3.2. Spike recovery

LPS spike recovery was assessed for all samples in the LAL, rFC, and MAT assays to verify the absence of assay interference. For all reported measurements, spike recoveries fell within the predefined acceptance range of 50–200%.

Across all tested samples, mean spike recoveries were 107% (SD =

Table 2

Overview of the 18 pAPI batches included in the study. The batches, representing 15 different phages, are grouped according to their bacterial host. Shown are titers determined by plaque assay (PFU/mL), endotoxin levels measured by LAL and rFC (EU/mL), and pyrogenicity assessed by MAT (EE/mL).

Gram-stain	Host	Phage	Titer (PFU/mL)	LAL (EU/mL)	rFC (EU/mL)	MAT (EE/mL)
Negative	<i>Pseudomonas aeruginosa</i>	PNM	4.70×10^9	< 5.0	< 5.0	< 17.0
		14/1	1.50×10^{10}	< 5.0	8.1	< 3.1
		PT07	8.00×10^9	224.6	247.1	< 6.3
		4P	3.00×10^9	< 5.0	< 5.0	< 3.1
		C	3.60×10^9	218.5	156.2	15.9
		DP1	7.60×10^9	< 5.0	< 5.0	< 3.1
		4K	2.60×10^9	< 5.0	< 5.0	< 3.1
		F	1.72×10^{10}	115.6	< 5.0	< 6.3
Negative	<i>Achromobacter xylosoxidans</i>	JWX	1.30×10^9	627.1	72.0	190.7
Negative	<i>Klebsiella pneumoniae</i>	NB2	4.30×10^9	247.4	240.2	> 2,500.0
		UZG11	2.50×10^9	65.4	6.2	135.3
Negative	<i>Morganella morganii</i>	266/267	2.80×10^9	> 50,000.0	46,255.4	> 12,500.0*
Positive	<i>Staphylococcus aureus</i>	ISP batch 1	4.50×10^{10}	16,306.5	3,764.2	172.8
		ISP batch 2	2.90×10^{10}	6,477.2	3,344.6	108.0
		ISP batch 3	4.00×10^{10}	382.0	93.7	31.9
		ISP batch 4	3.82×10^{10}	9.4	6.4	< 3.1
Positive	<i>Enterococcus faecalis</i>	KN	9.20×10^9	< 5.0	< 5.0	< 3.1
N/A	<i>Mycobacterium abscessus</i>	8UZL	1.60×10^{10}	< 5.0	< 5.0	< 3.1

*A subsequent experiment yielded a value of 15,246.2 EE/mL.

15%) for LAL, 96% (SD = 16%) for rFC, and 102% (SD = 29%) for MAT (Supplementary Table S1). Individual recoveries ranged between 80 and 127% (LAL), 53–112% (rFC), and 60–152% (MAT).

Where initial spike recovery fell outside the acceptance criteria, samples were further diluted and reassessed, and the highest valid dilution was used to determine the endotoxin (equivalent) level.

Similarly, NEP spike recoveries in MAT assays were all above 50%, meeting the predefined acceptance criterion for non-endotoxin pyrogens.

3.3. LAL vs. rFC for pAPIs

We compared the performance of the LAL and the rFC assay in

quantifying endotoxins in pAPIs (Table 2). The endotoxin concentrations obtained with the LAL assay ranged from less than 5 to more than 50,000 EU/mL, while rFC results ranged from less than 5 to 46,255 EU/mL. Both tests showed a strong correlation ($r = 0.9708$) and produced similar results for the majority of samples: 72% of samples (13/18) yielded LAL and rFC results within 50–200% of each other (Fig. 2). The remaining samples, 28% (5/18), exhibited higher LAL results compared to rFC.

3.4. MAT testing of pAPIs

The pAPIs were tested using the MAT with results expressed in EE/mL, assuming a dose–response curve similar to that of the standard

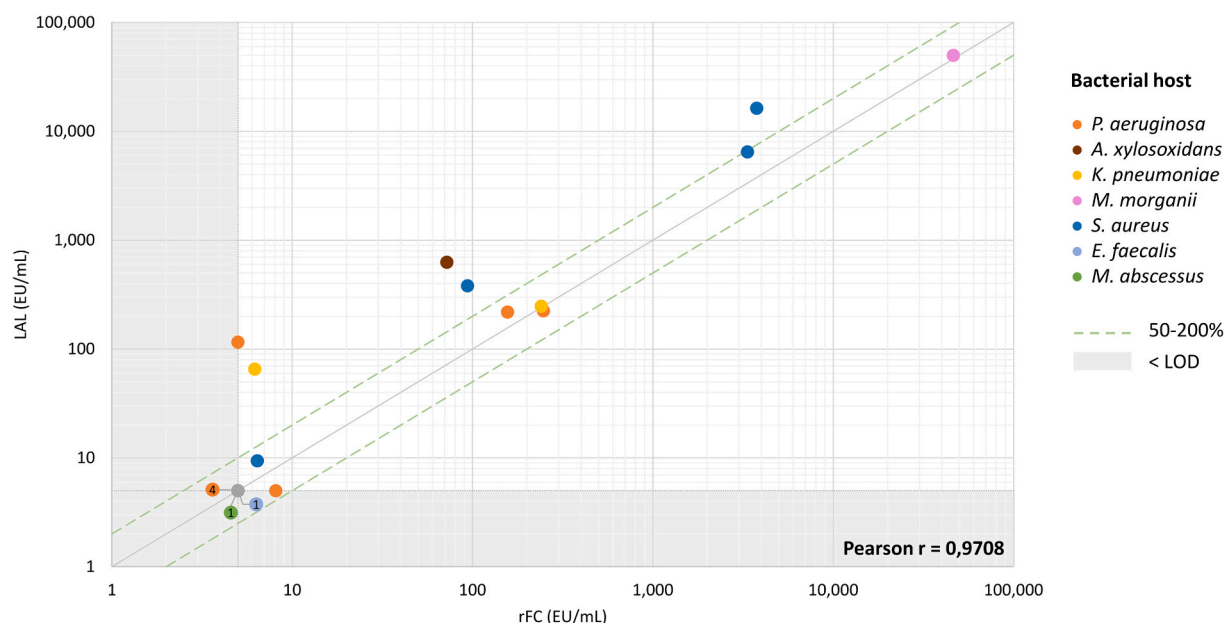


Fig. 2. Correlation between LAL and rFC assay results for 18 pAPI batches, plotted on log₁₀ axes. Each point represents a single batch, with colors indicating the bacterial host. The solid diagonal line denotes perfect agreement (100% equivalence), while the dashed lines represent the 50% and 200% equivalence thresholds. Shaded regions indicate values below the respective limits of detection (LOD = 5 EU/mL) for LAL and rFC. The grey marker represents all six batches that yielded results below the LOD for both assays: four batches of *P. aeruginosa* phages, one batch of a *M. abscessus* phage, and one batch of an *E. faecalis* phage. Their markers are slightly offset for clarity. Most batches fall within the 50–200% equivalence range, demonstrating strong concordance between the two assays (Pearson correlation = 0.9708). Batches outside this range consistently show higher endotoxin levels in LAL compared to rFC, consistent with the known susceptibility of LAL to matrix interference.

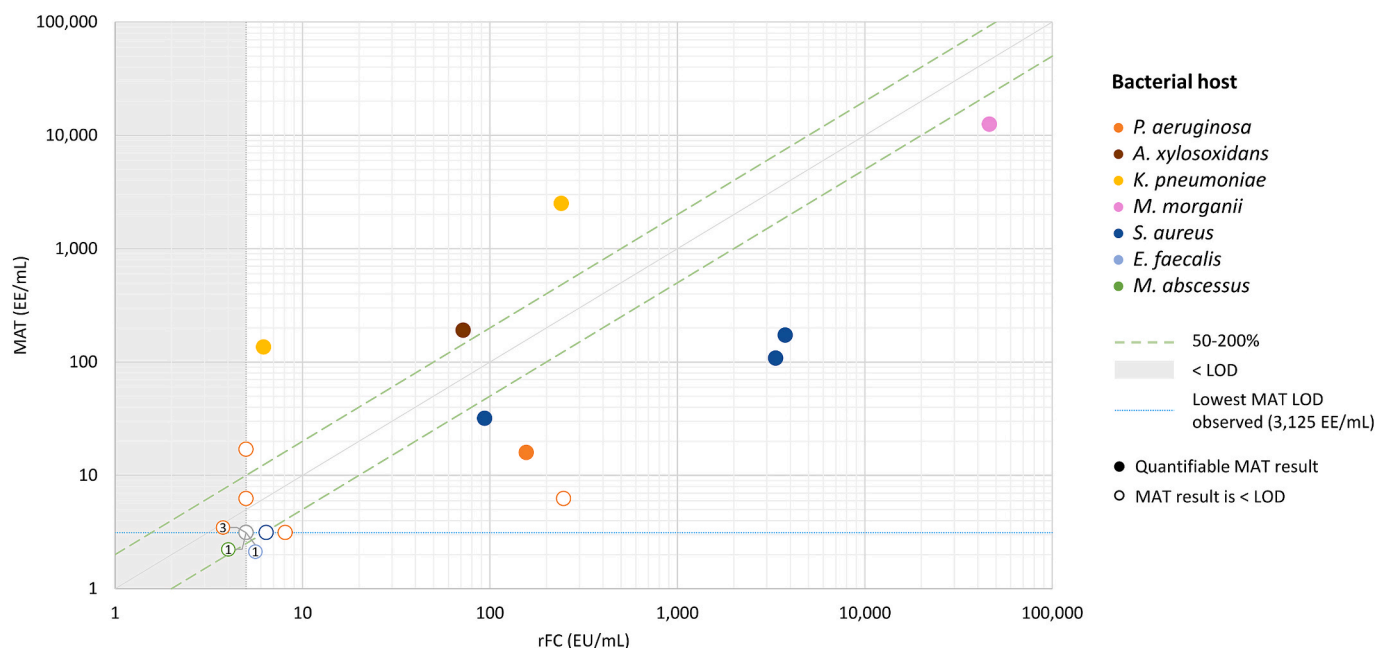


Fig. 3. Comparison of MAT and rFC assay results for 18 pAPI batches, plotted on $\log_{10}$ axes. Each point represents a single batch, with full markers indicating quantifiable MAT results and open markers representing samples with MAT results below the assay-specific limit of detection (LOD). Open markers are positioned at the intersection of the rFC value and the corresponding MAT LOD, which varies per test; the lowest observed MAT LOD (3.125 EE/mL) is indicated by a dotted horizontal line. The solid diagonal line represents 100% equivalence between MAT and rFC, with dashed lines marking the 50% and 200% equivalence thresholds. The shaded region denotes values below the fixed LOD of rFC (5 EU/mL). The grey marker represents all five batches that yielded results below the LOD for both assays: three batches of *P. aeruginosa* phages, one batch of a *M. abscessus* phage, and one batch of an *E. faecalis* phage. Their markers are slightly offset for clarity. Most data points fall below the 100% equivalence line, indicating lower pyrogenic activity detected by MAT compared to rFC. Only three samples showed higher MAT values than rFC.

E. coli LPS. Upon examination of the individual samples, MAT results were either consistent with rFC, representing all cases in which both assays were below the limit of detection (39% of cases, 7/18), or were lower than 50% of the corresponding rFC results (44% of cases, 8/18). Only a few samples (17%, 3/18) showed a MAT response exceeding 200% of the rFC value (see Fig. 3).

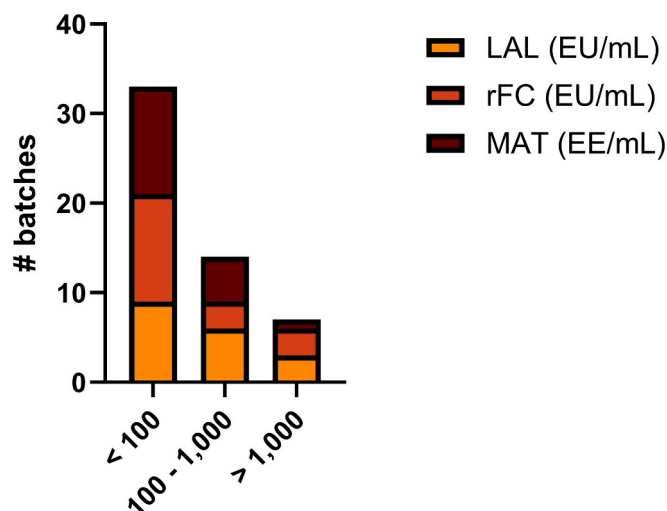


Fig. 4. Distribution of pyrogenic activity in 18 pAPI samples as assessed by LAL, rFC, and MAT assays. LAL testing revealed that 50% of samples contained less than 100 EU/mL, 33% ranged between 100 and 1,000 EU/mL, and 17% exceeded 1,000 EU/mL. In contrast, rFC and MAT assays showed lower overall pyrogenicity, with 67% of samples below 100 EE/mL in both assays. Intermediate pyrogenic activity (100 – 1,000 EE/mL) was more frequently observed in MAT (28%) than in rFC (17%), while high pyrogenic activity (> 1,000 EE/mL) was detected in only one sample by MAT, compared to three by rFC.

Of the 18 pAPI samples, LAL testing showed that 50% (9/18) contained less than 100 EU/mL, 33% (6/18) ranged between 100 and 1,000 EU/mL, and 17% (3/18) exceeded 1,000 EU/mL. In contrast, 67% (12/18) of samples demonstrated pyrogenic activity below 100 EE/mL in both the rFC and MAT assays. However, more samples exhibited intermediate pyrogenic activity (100–1,000 EE/mL) in MAT (28%, 5/18) than in rFC (17%, 3/18), and only one sample exhibited high pyrogenic activity (> 1,000 EE/mL) in MAT, compared to three samples in rFC. This distribution is visualized in Fig. 4. Overall, pyrogenicity measured with MAT was lower than that obtained with the endotoxin-specific assays.

4. Discussion

4.1. Comparison of LAL, rFC and MAT responses to pAPIs

When comparing LAL and rFC results, most samples yielded comparable values, within 50–200% of each other. For the few samples that differed, LAL consistently reported higher endotoxin levels than rFC. This aligns with the known characteristics of the assays: LAL relies on a multi-step enzymatic cascade that can be activated not only by endotoxin but also by β -glucans and other matrix components, whereas rFC uses a single recombinant Factor C enzyme, which reduces susceptibility to such interferences [38]. Therefore, the observed differences between LAL and rFC are not surprising.

Although β -glucan interference was not directly assessed in the present study, several aspects of the phage production and purification process could plausibly introduce β -glucan-like substances, including bacterial culture components, filtration materials, or residual host-derived polysaccharides. Future studies incorporating β -glucan-blocking reagents would be valuable to test this hypothesis.

In contrast to LAL, among the MAT results that deviated from the 50–200% range relative to rFC, the majority were lower than 50% of the

rFC value, rather than exceeding 200% (see Fig. 2). Given that the MAT detects pyrogenicity of both endotoxins and NEPs, while rFC only measures endotoxins, these results appear counterintuitive. Notably, samples exhibiting high to very high ($> 1,000$ EU/mL) endotoxin concentrations in LAL and rFC showed lower responses in MAT.

The lower MAT responses are unlikely to be due to direct cytotoxicity of the phage APIs on PBMCs, as bacteriophages are viruses that specifically infect bacterial hosts and cannot infect or replicate in eukaryotic cells. Although phages can interact with human immune cells, such interactions are not associated with loss of cell viability or cytotoxic effects [39,40]. Consistently, an MTS assay performed on PBMCs following exposure to pAPIs under MAT conditions confirmed metabolic activity comparable to that of untreated cells, with no reduction in viability observed (Supplementary Fig. S2).

Several factors may contribute to the discrepancy between MAT and LAL/rFC results. First, phage products may interfere with the crab-based endotoxin detection system, potentially leading to an overestimation of endotoxin concentration. Second, phages themselves may have an anti-inflammatory or immunomodulatory effect in MAT, an aspect that will be addressed further in this discussion. Third, and importantly, LPS derived from non-*E. coli* species, such as *Klebsiella pneumoniae* and *Pseudomonas aeruginosa*, may exhibit differential reactivity in LAL and rFC assays compared to *E. coli* LPS [41].

LAL and rFC assays rely on an enzymatic amplification cascade initiated by the interaction of Factor C with the Lipid A moiety of LPS. This biochemical amplification confers high analytical sensitivity and reflects the structural recognition of Lipid A, often resulting in a conservative estimation of endotoxin burden. In contrast, the MAT measures functional pyrogenicity, i.e., the ability of contaminants to activate human monocytes and induce cytokine release.

It is well established that LPS derived from different Gram-negative bacterial species show marked variability in their biological potency [42]. Structural differences in Lipid A can strongly influence the biological activity of LPS, without proportionally affecting Factor C activation. Because endotoxin assays are typically calibrated against *E. coli* LPS, such species-dependent differences can lead to an apparent over- or underestimation of biologically relevant pyrogenicity when assessed solely by enzymatic assays.

Consequently, the lower MAT responses observed for samples with high endotoxin levels measured by LAL and rFC do not necessarily indicate assay interference or analytical artefacts, but rather reflect differences between enzymatic sensitivity and human biological potency. This highlights the complementary nature of endotoxin-specific assays and MAT, particularly for complex biological products such as pAPIs, where endotoxin composition and origin may differ substantially from the calibration standards used in traditional assays.

One sample, *K. pneumoniae* phage NB2, exhibits a significantly larger response in MAT ($> 2,500.0$ EE/mL) compared to LAL and rFC (247.4 and 240.2 EU/mL), representing an outlier compared to the overall trend observed in this study. This difference may be linked to the instability of this particular batch. Notably, a white precipitate was observed in the sample prior to pyrogen testing (shown in Supplementary Fig. S3), suggesting possible phage aggregation and reduced sample homogeneity. Such aggregation could have led to uneven exposure of PBMCs to biologically active components, potentially resulting in an amplified MAT response. In addition, aggregated phage material or copurified host-derived components may enhance cellular immune activation, whereas enzyme-based assays such as LAL and rFC are less sensitive to particulate or aggregated structures.

Interestingly, two batches of phage ISP targeting *Staphylococcus aureus* exhibited a moderate MAT response and unusually high endotoxin concentrations in both LAL and rFC assays, confirmed by multiple tests. This finding is unforeseen, as the presence of endotoxins is not expected in pAPIs produced on Gram-positive bacteria. Moreover, six previously produced ISP batches consistently showed low endotoxin levels when analyzed by LAL (see Supplementary Table S2). The

underlying cause of this anomaly remains unclear and warrants further investigation. LPS contamination originating from water or culture media is considered unlikely, as the affected batches were subjected to an affinity-based endotoxin removal step, and a second treatment did not reduce LAL or rFC signals. Possible explanations include matrix components, host-derived factors, or particulate matter artificially activating Factor C-based assays without triggering a corresponding pro-inflammatory response in human immune cells.

To minimize potential risks to patient safety and prevent the loss of production efforts, a novel *S. aureus* strain was selected by the pAPI producers (QAMH) as the production host for ISP. The three batches subsequently obtained with this strain showed no detectable endotoxin activity in rFC assays (see Supplementary Table S2), supporting the hypothesis that host-derived factors were responsible for the observed assay interference. These findings underscore the importance of host strain selection and the use of orthogonal, human-relevant pyrogen testing strategies during pAPI development and batch release, and highlight a critical limitation of LAL- and rFC-based assays for complex biopharmaceutics.

4.2. Establishment of a maximum valid dilution (MVD) for pAPIs

The MVD, as defined by the Ph. Eur. [33,36,37], represents the greatest dilution factor at which the LAL, rFC or MAT can be performed without compromising its validity. While the specific terminology for certain parameters may vary slightly between the endotoxin tests and the MAT, the underlying concepts are consistent across all three assays. For the MAT, the MVD calculation requires three parameters: the Contaminant Limit Concentration (CLC), the product concentration (C), and the test sensitivity. The CLC is calculated as the threshold pyrogenic dose per kilogram of body mass (K-value) per maximum recommended bolus dose of product per kilogram of body mass (M) (see Equation 1). K-values are expressed in endotoxin equivalents (EE) and depend on the route of administration. For intravenous administration, the threshold pyrogenic dose is 5 EE per kilogram of body mass, whereas for parenteral formulations administered per square meter of body surface, the limit is 100 EE per m². The maximum bolus dose per kilogram of body mass (M) can be expressed in mass, units, or volume per kilogram (and per hour, if infused continuously), and must be established for each product.

For phage therapeutics, the maximum bolus dose is currently undefined. Treatments are not yet standardized, and dosage is highly dependent on the nature and localization of the infection. Additionally, clinical case reports typically describe only the phage concentration used (in plaque-forming units per milliliter (PFU/mL)), but rarely provide the total volume administered [11,43]. In practice, the dosage of phage therapeutics is often adjusted based on the measured endotoxin concentration, and products are diluted until the amount of endotoxins falls within acceptable limits. Unlike conventional medicines, phages have no theoretical upper dose limit, as no toxic dose has been identified to date [44]. Although higher phage titers do not necessarily correlate with improved clinical outcome [11], some centers favor the use of highly concentrated preparations. Consequently, it is not currently possible to define a universal maximum bolus dose for phage therapeutics, which complicates the calculation of both CLC and MVD.

With a number of conservative assumptions, we will attempt a theoretical estimation. Assuming a phage therapeutic is administered intravenously (K-value for intravenous administration = 5 EE/kg), with a bolus dose of 20 mL (irrespective of the phage concentration), to an adult of 60 kg, the CLC of the final phage formulation would be equal to 15 EE/mL. Since pAPIs, the raw material for preparation of phage therapeutic products, usually contain 10^9 - 10^{10} PFU/mL, and clinical protocols in Belgium often work with a final phage concentration of 10^7 PFU/mL, we will assume that pAPIs will be diluted at least 100-fold. Therefore, a pAPI destined for an adult of 60 kg could contain up to 1,500 EE/mL, whereas this limit would drop to 75 EE/mL for a newborn

of 3 kg. In case the pAPI is administered undiluted, this permissible endotoxin level would be substantially reduced. The MVD is calculated by multiplying the CLC by the concentration of the product and dividing it by the test sensitivity (see Equation 2). Here, the actual product concentration (C) becomes irrelevant, as the CLC has already been normalized per mL of phage product. The test sensitivity of the MAT is defined as the lowest endotoxin reference standard concentration on the standard curve whose response exceeds a cut-off value, which is calculated from the mean and standard deviation of the blank responses (see Equation 3). As it corresponds to an actual point on the standard curve, the sensitivity must be confirmed for each test. Taking 0.03125 EE/mL as the test sensitivity of the MAT, i.e., the best sensitivity observed in our experiments (lowest detectable concentration), the MVD would correspond to 48,000 for adults and 2,400 for children. A lower assay sensitivity (a higher minimum detectable concentration) would proportionally reduce the acceptable dilution.

Equation 1 – contaminant limit concentration. K = threshold pyrogenic dose per kilogram of body mass; M = maximum recommended bolus dose of product per kilogram of body mass.

$$CLC = \frac{K}{M}$$

Equation 2 – Maximum valid dilution. CLC = contaminant limit concentration; C = product concentration; Test sensitivity = lowest endotoxin reference standard concentration on the standard curve whose response exceeds the cut-off value (see Equation 3).

$$MVD = \frac{CLC \cdot C}{\text{Test sensitivity}}$$

Equation 3 – Cut-off value. $\bar{x}$ = mean of the responses obtained for the 4 blank replicates; s = standard deviation of the responses obtained for the 4 blank replicates.

$$\text{Cut-off value} = \bar{x} + 3s$$

Applying this CLC, which is identical across the LAL, rFC, and MAT methods, 15 out of 18 samples would pass the pyrogenicity test for use on adults based on LAL and rFC results, while only 9 (LAL) and 11 (rFC) would qualify for pediatric use. In contrast, MAT classified 12 out of 18 batches as suitable for children and 16 out of 18 for adults. Thus, based on this dataset, it seems using the MAT as a reference would reduce the likelihood of unnecessarily discarding batches.

Naturally, these estimates reflect a stringent scenario, involving a relatively large intravenous bolus of phage product. In practice, phage therapeutics are frequently administered via other routes (e.g., nebulization, topical application, catheter instillation [10]), where pyrogenicity poses a lower risk. Moreover, many phage therapies are still delivered under compassionate use frameworks, where the treating physician may weigh potential pyrogenic risks against clinical necessity. Nonetheless, the possibility of intravenous delivery of highly concentrated phage therapeutics, as practiced in some centers [43], underscores the importance of including rigorous pyrogenicity testing in the quality control of pAPIs.

4.3. Advantages and disadvantages of MAT compared to endotoxin tests for QC of pAPIs

In addition to detecting pyrogens missed by endotoxin tests but still causing adverse events in patients [43], the MAT can provide important contextual information when elevated endotoxin results are observed in products with low apparent clinical risk (e.g., *Staphylococcus* phage ISP) and help prevent unnecessary batch rejection. However, MAT has costs and handling disadvantages. Commercial MAT kits cost about twice the price of LAL or rFC kits, and Pharmacopoeial guidelines [37] limit MAT to two samples per test, compared with 20 samples per plate for LAL and rFC, making it roughly 20 times more expensive per sample. The

conventional cytokine-release assay is also slower, requiring overnight incubation, and has a narrow dynamic range, which complicates the analysis of unknown samples, often necessitating repeated testing or parallel dilution series. Some of these limitations, particularly assay duration, may be mitigated by more recently developed MAT variants based on NF- κ B reporter cell lines, which allow assay completion within a single working day.

Despite its drawbacks, we are convinced MAT represents a valuable component of the QC process of pAPIs. While not ideal for in-process control, it is crucially informative for batch release of new productions for which the presence of NEPs cannot be ruled out. Indeed, this study demonstrates that endotoxin testing alone captures only part of the pyrogenicity picture, especially in complex biological products such as pAPIs, where pyrogenic components can be of diverse nature and origin. While endotoxin assays remain valuable for rapid, cost-effective in-process control, they don't provide complete pyrogenic risk assessment. We therefore propose a two-level QC approach, using rFC initially to provide a provisional endotoxin value, guiding MAT dilution selection and minimizing repeat testing. We deliberately favor rFC over LAL, as rFC has been shown to deliver equivalent or superior performance [38,45] while avoiding the use of horseshoe crab blood, thereby aligning with the 3Rs principles. In this framework, MAT serves as a complementary assay to detect NEPs and to investigate borderline or unexpected rFC results. Even when MAT indicates low biological activity, a high chemical burden of LPS detected by rFC may still carry a clinical risk, particularly for vulnerable patient populations (e.g., immunocompromised patients), highlighting the need to carefully balance the urgency of providing life-saving therapy with the imperative of patient safety. This strategy leverages the speed and sensitivity of rFC with the broader safety scope of MAT, offering a more complete assessment of pAPIs interactions with the human immune system.

4.4. Phage-monocyte interaction: Misleading signals or immune insights?

Phages are abundant in and on the human body, particularly in the gut, and are recognized as integral components of the microbiome. Given their ubiquitous presence and their inability to infect human cells, they are considered generally safe. This claim is also supported by the many clinical case reports of phage therapy, expressing no safety concerns [44,46–48]. However, evidence suggests that, although they can't replicate in, they can enter mammalian cells and elicit immune responses [49–51]. It has been proposed that phages, or their structural components, may be recognized by pattern recognition receptors, such as the transmembrane Toll-Like Receptors (TLRs), similarly to eukaryotic viruses [52]. TLRs play a crucial role in the initiation of innate immune responses by triggering a signaling cascade, resulting in cytokine production and release.

To date, observed immune responses to phages are heterogeneous and lack consistent patterns across different studies [53], probably partly because purification is technically challenging and contaminants may confound results. In some studies, phages or their components induce secretion of pro-inflammatory cytokines (e.g., IL-6, TNF- α , IL-1 β) [54,55], while in others they produce anti-inflammatory signals (e.g., interleukin-10 (IL-10) and transforming growth factor β (TGF- β)) [55,56]. The nature of the immune response appears to be influenced by multiple factors, including phage type, dosage, route of administration, purification level, and the immunological status of the host. Therefore, findings from one phage cannot be extrapolated to others, underscoring the need for phage-specific investigations.

The interaction between phages and monocytes presents a potential challenge for the interpretation of results in the MAT, as it may increase the likelihood of false positives, false negatives, or ambiguous outcomes. However, we argue that this aspect precisely highlights the relevance of the MAT. As a human cell-based assay, MAT is uniquely positioned to capture the complex immune responses that are likely to occur in patients, thereby providing a more realistic assessment of product safety

than endotoxin testing. Furthermore, potential interference from the product matrix is systematically assessed by evaluating the recovery of both LPS and NEP spikes for each individual product. In all our experiments, recovery of LPS and NEP spikes met acceptable criteria, confirming the robustness of the assay under the tested conditions. Additionally, complementing IL-6-based pyrogenicity testing with multiplex cytokine profiling could further strengthen the MAT by capturing broader aspects of monocyte activation and providing a more detailed view of immune response patterns.

Undoubtedly, questions regarding the interaction between specific bacteriophages and human monocytes, and their implications for the MAT, remain unresolved and warrant further investigation. Nonetheless, the findings presented here offer an initial glimpse into the pyrogenic profile of pAPIs, as perceived by the human immune system. Importantly, this approach avoids reliance on animal-based assays, aligning with the principles of the 3Rs and supporting the development of more ethical and human-relevant safety testing strategies.

4.5. Study limitations

This study has several limitations that should be acknowledged when interpreting the results and their implications for phage therapy quality control.

The sample size is relatively limited, with 18 phage batches representing only seven bacterial species. Phage therapy encompasses a much broader range of target bacteria, and the pyrogenic profiles of phages propagated on other clinically relevant species remain to be characterized. Additionally, the limited number of batches per phage type restricts our ability to assess batch-to-batch variability comprehensively.

Sample stability and homogeneity were not systematically controlled or quantified. The isolated outlier case of *K. pneumoniae* phage NB2, which exhibited a markedly elevated MAT response ($>2,500.0$ EE/mL) compared to rFC and LAL results, underscores that MAT results may be influenced by the physical state of the sample. Future studies should include systematic evaluation of sample stability parameters (e.g., particle size distribution, aggregation state) and their potential impact on MAT outcomes, as well as establish clear acceptance criteria for sample physical quality before testing.

Glucan-blocking buffers were not used in the LAL assay, and therefore β -glucan interference cannot be excluded as a contributing factor to the higher LAL values observed. Additionally, spike recovery was assessed only in diluted samples. Spiking in undiluted samples could provide further insight into matrix interference effects under neat conditions.

Finally, this study focused on purified phage API rather than final formulated drug products. The pyrogenic profile may change during formulation, and additional excipients or stabilizers could introduce new sources of pyrogenic contamination or matrix interference. The applicability of our findings to final pharmaceutical products requires further investigation.

Despite these limitations, this study provides valuable insights into the pyrogenic testing landscape for phage therapy and establishes a foundation for more comprehensive QC strategies that balance safety, efficiency, and regulatory compliance.

5. Conclusion

This study highlights the complexity of assessing pyrogenicity in pAPIs and the limitations of relying solely on biochemical endotoxin assays. While LAL and rFC often yield comparable results, LAL is less specific and can overestimate endotoxin levels. In contrast, the MAT offers a more biologically relevant assessment by capturing the immune response of human monocytes, making it particularly valuable for detecting risks associated with NEPs and for clarifying borderline or unexpected endotoxin test results.

The majority of pAPIs tested were pyrogenically safe at the time of

testing, and none showed evidence of NEP contamination. However, discordant results observed between the biochemical and biological assays raise some regulatory and safety concerns. High endotoxin chemical loads detected by rFC represent a real patient risk that cannot be dismissed solely based on MAT outcomes. At the same time, batches that do not show high biological activity may still be considered for use in a risk–benefit context, particularly for patients requiring urgent treatment.

Based on our findings, we propose a two-tiered QC strategy using rFC for initial screening and MAT as a complement, combining efficiency with a more comprehensive safety profile. This approach not only enhances patient safety but also aligns with ethical and regulatory best practices by supporting the 3Rs principles and advancing human-relevant testing in phage therapy development.

Finally, phages themselves may have immunomodulatory effects that could influence MAT outcomes, underscoring the need for further research into phage–host immune interactions. In the meantime, systematic matrix interference testing ensures the reliability of MAT results for pAPI quality control.

6. Declaration of generative AI and AI-assisted technologies in the manuscript preparation process:

ChatGPT, Claude AI and Le Chat (Mistral AI) were used to enhance the clarity and coherence of the writing. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Marie Willocx: Writing – original draft, Visualization, Validation, Investigation, Conceptualization. **Jean-Paul Pirnay:** Writing – review & editing, Resources. **Maia Merabishvili:** Writing – review & editing, Resources. **Flore Laurent:** Writing – review & editing, Validation, Investigation. **Sophie Clouet:** Writing – review & editing, Investigation. **Rob Lavigne:** Writing – review & editing, Supervision, Conceptualization. **Pieter-Jan Ceyssens:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Celine Vanhee:** Writing – original draft, Supervision, Resources, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ejpb.2026.115068>.

Data availability

Data will be made available on request.

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